

The 2048 Chrysopoeia: An Explicit Construction Program for the $d = 2048$ SIC-POVM Moduli

C. Lando Mills

Independent Researcher

Correspondence: c.landonmills@gmail.com

ORCID: 0000-0003-0003-0552

Declarations

Availability of data and material

The companion formalization and its Lean modules live in the `p4ramill` library of `p4rakernel` (<https://github.com/umpolungfish/p4rakernel>), frozen at commit `eea2c0c` on branch `crystalline/manuscripts3-2026-07-07` (tag `crystalline-manuscripts3-v1`). Field computations reported in the manuscript were performed in PARI/GP. This manuscript trio is frozen in `ig-docs` (<https://github.com/umpolungfish/ig-docs>) on the same branch name and tag.

Competing interests

The author declares no competing interests.

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Authors' contributions

C. Lando Mills: conceptualization, formalization, investigation, validation, writing (original draft), writing (review and editing).

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The author thanks Harry T. Larson for imparting the importance of catching rising problems and never letting them go. The $d = 2048$ moduli program was undertaken in the same spirit: when unconstrained recovery stalled at a spurious

local minimum, the route through Stark units, the ray class tower, and the integer norm sieve was not dropped. Field computations were performed in PARI/GP; the companion formalization is the SIC–Stark–12th manuscript in this trio.